

MATSUE, Japan

WMO: 477410

Lat: 35.4572N

Lon: 133.0656E

Elev: 22

StdP: 101.06

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	-0.2	-6.5	2.2	2.8	-5.2	2.4	3.3	13.8	1.9	12.7	2.4	3.8	270	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	33.5	25.8	32.3	25.6	31.1	25.2	26.6	31.3	26.1	30.7	25.6	29.9	4.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	29.0	24.8	19.9	28.6	24.4	19.3	28.3	83.3	31.4	81.2	30.8	79.1	30.0	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.8	8.5	DB	-3.3	35.5	1.3	1.0	-4.2	36.3	-5.0	36.9	-5.7	37.5	-6.7	38.2	(4)
(5)			WB	-4.0	27.5	1.3	0.5	-5.0	27.9	-5.8	28.2	-6.5	28.5	-7.5	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.5	4.6	5.1	8.2	13.3	18.3	22.0	26.3	27.7	23.2	17.7	12.2	7.1	(6)	
	DBStd	8.33	2.34	2.66	3.11	3.23	2.56	2.28	2.54	2.22	2.73	2.69	2.91	2.72	(7)	
	HDD10.0	496	168	139	75	8	0	0	0	0	0	0	11	96	(8)	
	HDD18.3	1874	426	370	315	153	32	1	0	0	1	43	186	348	(9)	
	CDD10.0	2517	1	2	18	107	257	361	505	548	397	239	76	6	(10)	
	CDD18.3	854	0	0	0	2	32	112	247	290	147	24	1	0	(11)	
	CDH23.3	6905	0	0	1	17	168	518	2241	3025	867	68	1	0	(12)	
	CDH26.7	2309	0	0	0	1	18	94	785	1191	212	8	0	0	(13)	
(14) Wind	WSAvg	3.3	3.8	3.5	3.6	3.6	3.4	3.1	3.4	3.1	2.9	2.7	3.0	3.7	(14)	
(15) Precipitation	PrecAvg	1774	163	116	130	109	140	166	230	131	211	131	129	155	(15)	
	PrecMax	2225	243	200	214	223	376	373	597	295	482	358	227	270	(16)	
	PrecMin	1476	96	71	62	39	30	26	8	23	51	48	35	38	(17)	
	PrecStd	206	40	34	41	45	84	94	155	75	117	84	50	57	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.5	20.9	25.5	28.7	30.8	34.2	35.5	32.8	27.6	22.2	16.5	(19)	
		MCWB	10.5	12.4	13.3	16.6	19.2	22.9	25.9	25.9	25.3	21.9	16.5	12.1	(20)	
	2%	DB	11.3	13.6	17.9	23.0	26.8	29.0	32.9	34.1	30.7	25.1	20.0	14.3	(21)	
		MCWB	7.7	9.6	12.0	15.7	18.1	22.0	25.8	25.8	24.5	20.0	15.4	10.7	(22)	
	5%	DB	9.8	11.5	15.8	21.1	25.2	27.6	31.7	33.0	29.1	23.5	18.6	12.9	(23)	
		MCWB	6.8	8.1	11.0	14.6	17.6	21.7	25.5	25.8	23.7	18.8	14.8	9.7	(24)	
	10%	DB	8.6	9.9	13.9	19.3	23.7	26.3	30.6	31.8	27.5	22.2	17.2	11.6	(25)	
		MCWB	5.8	6.9	10.0	13.7	16.9	21.2	25.1	25.5	23.0	17.9	13.9	8.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	13.4	14.8	19.2	21.4	24.9	27.1	27.5	26.2	22.9	18.3	13.7	(27)	
		MCDB	13.4	16.8	18.8	22.7	24.5	28.8	31.9	32.5	30.5	26.3	20.3	14.7	(28)	
	2%	WB	8.6	10.6	13.1	17.0	20.0	23.9	26.4	26.8	25.2	21.5	16.8	11.6	(29)	
		MCDB	10.4	12.8	16.1	21.2	23.2	26.9	31.1	31.6	29.2	24.0	18.8	13.5	(30)	
	5%	WB	7.2	8.7	11.7	15.6	19.0	23.0	25.9	26.3	24.5	20.1	15.6	10.2	(31)	
		MCDB	9.2	10.9	14.7	19.6	23.2	26.0	30.4	31.0	28.0	22.3	17.9	12.2	(32)	
	10%	WB	6.1	7.2	10.4	14.6	18.1	22.1	25.5	25.9	23.8	18.7	14.3	9.0	(33)	
		MCDB	8.2	9.4	13.5	18.0	22.1	25.0	29.6	30.4	26.7	21.2	16.7	11.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.1	8.6	9.3	8.9	7.1	6.3	7.0	6.7	7.7	7.6	6.4	(35)	
		MCDBR	8.7	11.0	12.1	12.3	11.8	9.3	8.2	8.6	8.3	9.1	9.7	9.2	(36)	
	5% WB	MCWBR	5.8	7.3	7.4	6.2	5.3	4.0	3.3	3.1	3.5	4.6	5.8	6.0	(37)	
		MCWBR	8.1	9.6	10.3	9.9	9.0	6.8	7.3	7.6	7.2	7.0	8.2	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.436	0.471	0.539	0.557	0.554	0.572	0.569	0.531	0.490	0.451	0.434	0.428	(40)	
		taud	2.003	1.944	1.809	1.820	1.885	1.920	1.987	2.097	2.162	2.157	2.110	2.048	(41)	
	Ebn at Noon		720	746	731	746	755	740	738	759	771	762	723	700	(42)	
		Edh at Noon	136	162	202	209	199	192	178	157	140	129	122	123	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.73	2.47	3.63	4.78	5.28	4.94	4.91	5.09	3.94	3.21	2.19	1.57	(44)	
	RadStd		0.14	0.24	0.32	0.39	0.51	0.38	0.73	0.56	0.34	0.28	0.18	0.13	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	+0.49	N/A	N/A	N/A	N/A	N/A	N/A
Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page